Databases – Exercises SQL

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1. Consider the following relational database:

   Person($\underline{\text{id}}$, name, city, age)

   Knows($\underline{\text{id1}} \rightarrow$ Person, $\underline{\text{id2}} \rightarrow$ Person)

   Likes($\underline{\text{id1}} \rightarrow$ Person, $\underline{\text{id2}} \rightarrow$ Person)

   Friend($\underline{\text{id1}} \rightarrow$ Person, $\underline{\text{id2}} \rightarrow$ Person)

   Account($\underline{\text{accountNr}}$, balance)

   BelongsTo($\underline{\text{id}} \rightarrow$ Person, $\underline{\text{accountNr}} \rightarrow$ Account)

   We assume that the *friends* relation is **symmetric**. If (A, B) is a tuple in the Friend table, then A is friend of B and B is friend of A. We do *not* assume that the relations *knows* and *likes* are symmetric. If (A, B) is a tuple in the Knows table, then A knows B, but B does not necessarily know A.

   Give an expression in SQL to express each of the following queries:

   (a) Find the name of all persons that are younger than 40.

   (b) Find the name of all persons that have a friend.
      i. Will the query return duplicate results?
      ii. Does adding distinct solve the problem?
      iii. Improve the query by using `in`.
      iv. Do the same with `exists`.

   (c) Find the name of all people that have no friends using `not in`.

   (d) Find the name of all people that have no friends using `not exits`.

   (e) Find the name of all persons that have a bank account that contains more than 1000 euros.
      i. Does the above query return duplicates?
      ii. Does adding distinct solve the problem?
      iii. Do the query using `in`.

   (f) Find all people that know at least two people. Do not use `group by`.

   (g) Find all people that know at least two people using `group by`

2. Consider the following relational database:

   Employee(<u>employeeName</u>, street, city)

   Works(<u>employeeName</u> $\rightarrow$ Employee, companyName $\rightarrow$ Company, salary)

   Company(<u>companyName</u>, city)

   Manages(<u>employeeName</u> $\rightarrow$ Employee, <u>managerName</u> $\rightarrow$ Employee)

   Give an expression in SQL to express each of the following queries:

   (a) Find the names and the cities of residence of all employees who work for the "First Bank Corporation".

   (b) Find the names, street addresses, and cities of residence of all employees who work for "First Bank Corporation" and earn more than $10,000.

   (c) Find all employees in the database who do not work for "First Bank Corporation".

   (d) Find all employees in the database who earn more than each employee of "Small Bank Corporation".

   (e) Assume that the companies may be located in several cities. Find all companies located only in the cities in which the "Small Bank Corporation" (SBC) is located.

   For example, assume SBC is located in Amsterdam and Eindhoven. Then we want all the companies that are located in any of these two cities or in both.